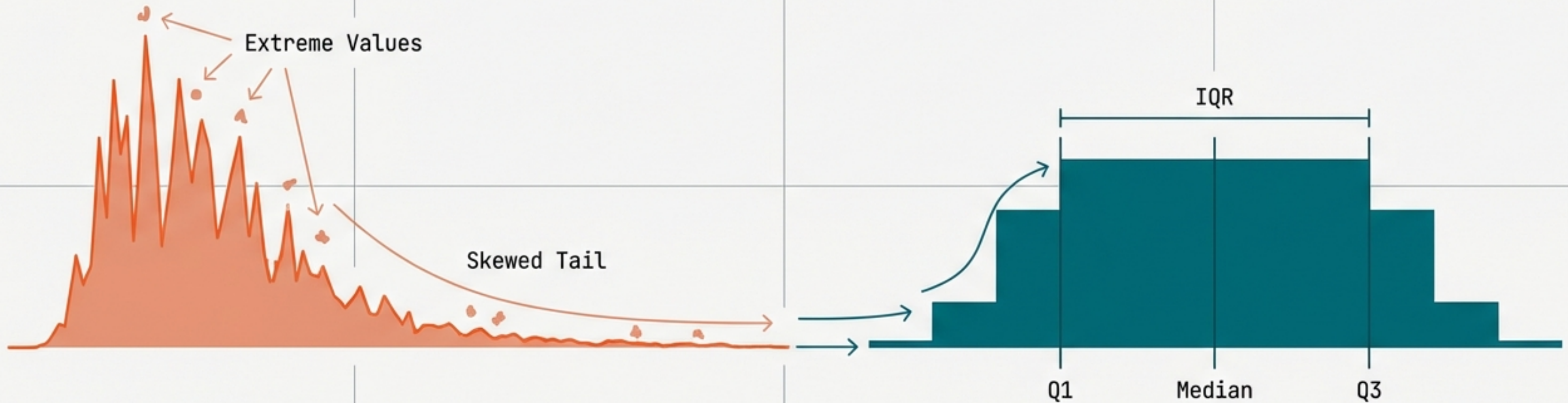


Taming Skewed Data

Outlier Detection & Removal: Mastering the Interquartile Range (IQR) Method

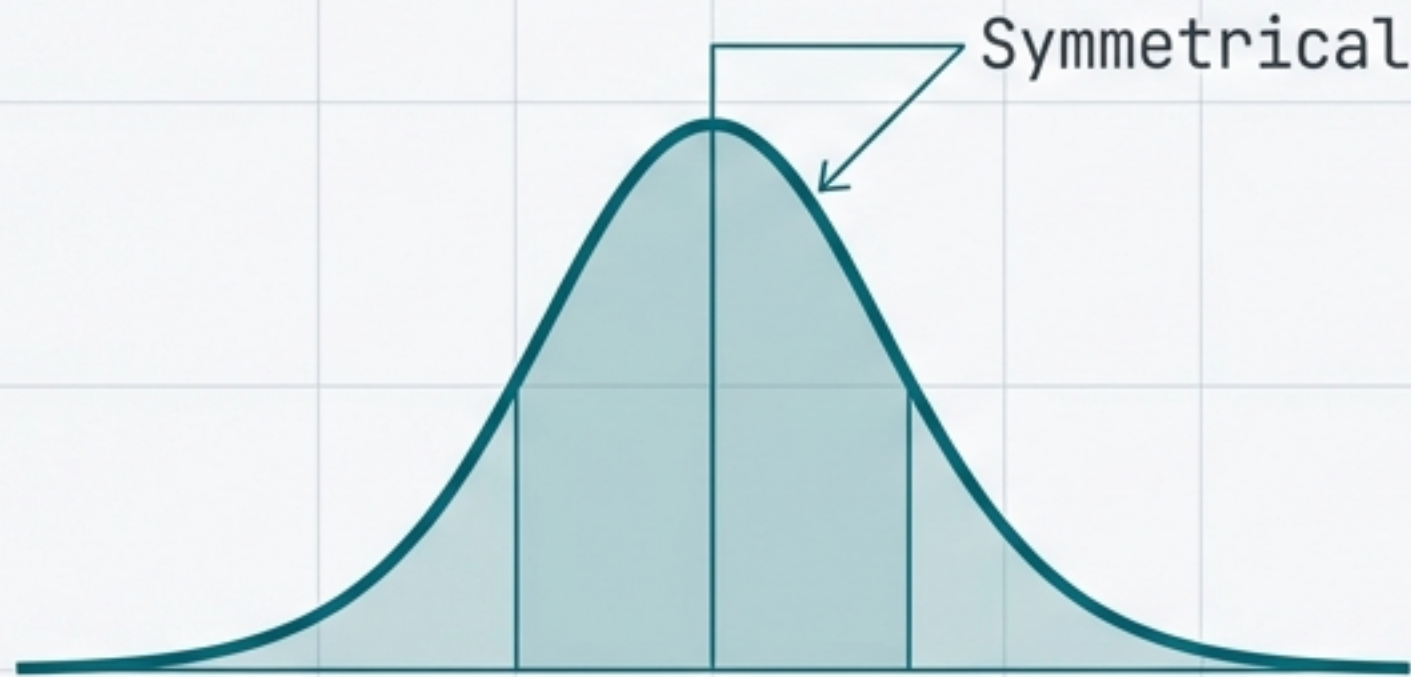


WORKFLOW OVERVIEW:

1. Diagnosis: Identifying Skewed Distributions
2. Prescription: Mathematical Boundary Calculation
3. Treatment: Pythonic Implementation (Trimming vs. Capping)

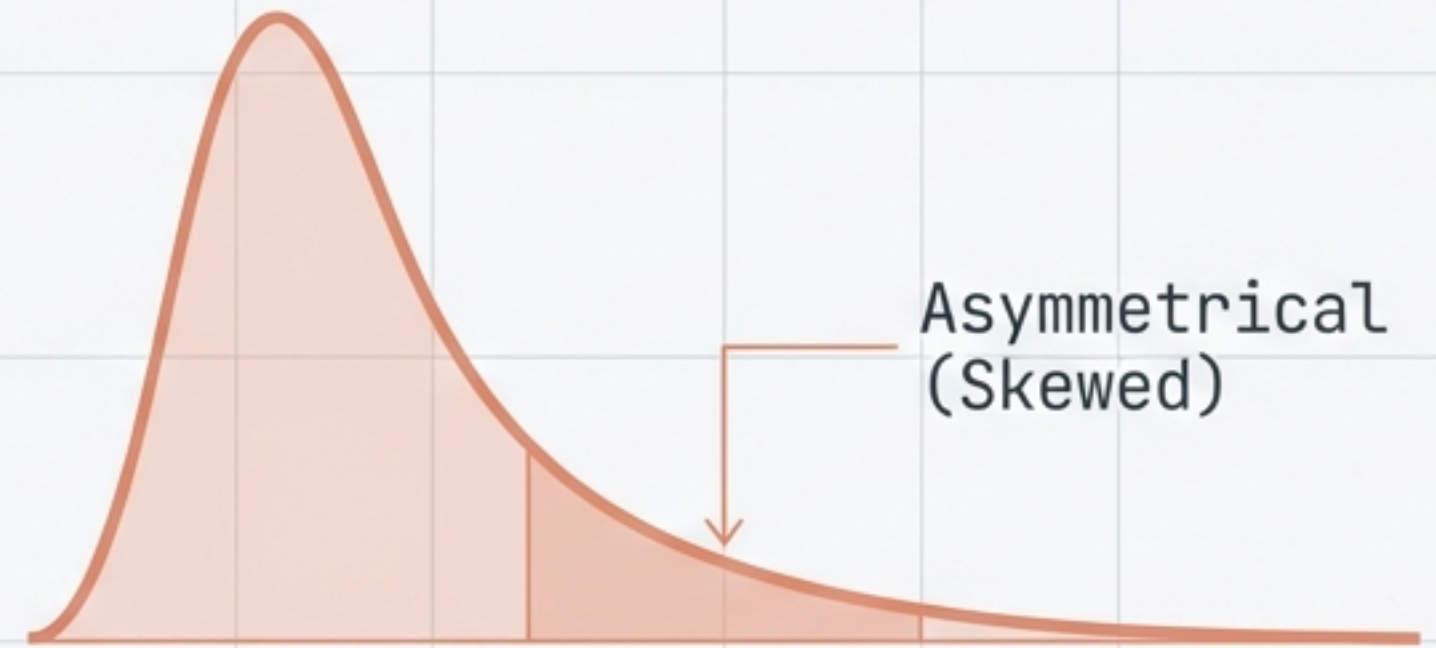
The Prerequisite: Diagnosis Before Treatment

Normal Distribution



Use Z-Score Method

Skewed Distribution

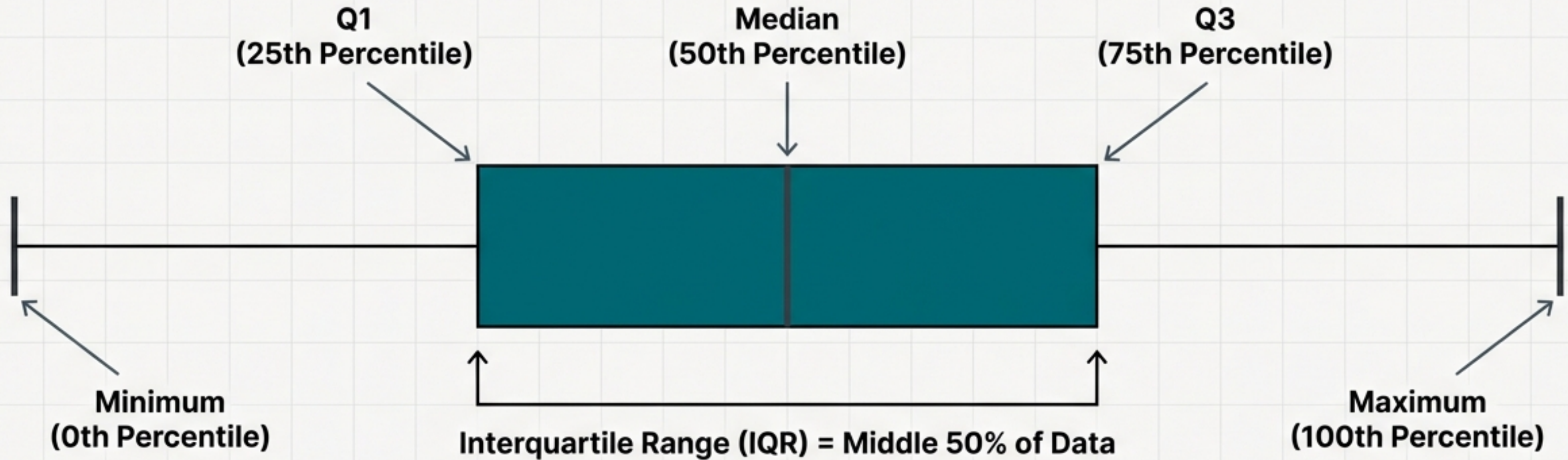


Use IQR Method

NOTE

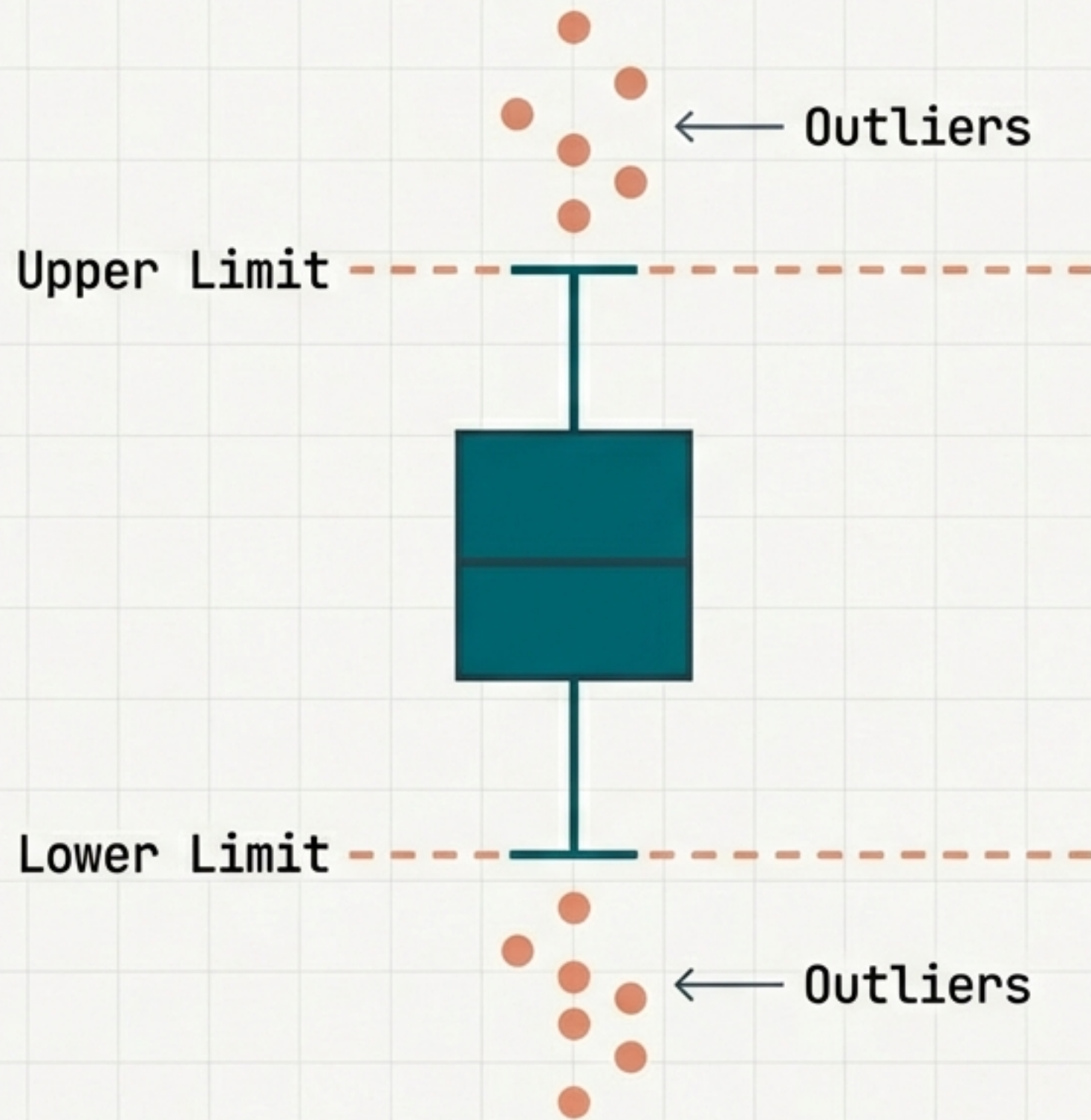
If a feature is not normally distributed, IQR is the standard approach for outlier detection.

The Toolkit: Anatomy of a Box Plot



Key Concept: The IQR is the distance between the 75th and 25th percentiles. It represents the bulk of the valid data.

The Prescription: The IQR Proximity Rule



Step 1: Calculate Range

$$\text{IQR} = Q3 - Q1$$

Step 2: Define Boundaries

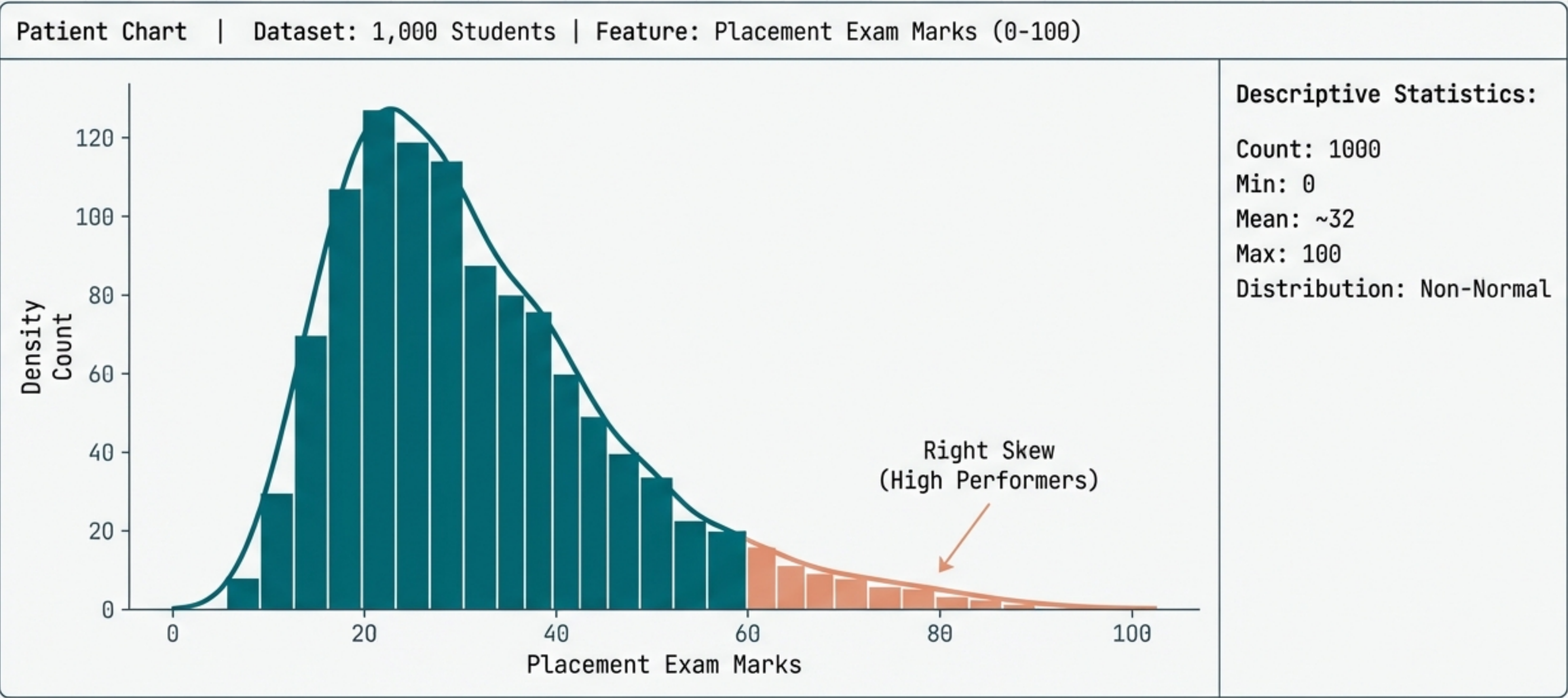
$$\text{Upper Limit} = Q3 + (1.5 \times \text{IQR})$$

$$\text{Lower Limit} = Q1 - (1.5 \times \text{IQR})$$

The Rule:

Any data point $>$ Upper Limit or $<$ Lower Limit is a statistical outlier.

Case Study: Student Placement Data



Unlike the CGPA column (which was normal), the Exam Marks show a heavy skew. This is our target for IQR treatment.

Calculating the Boundaries

Find Quartiles

$$Q1 (25\%) = 17$$

$$Q3 (75\%) = 44$$

Calculate IQR

$$IQR = 44 - 17$$

$$IQR = 27$$

Set Limits

$$\text{Upper Limit} = 44 + (1.5 \times 27) = 84.5$$

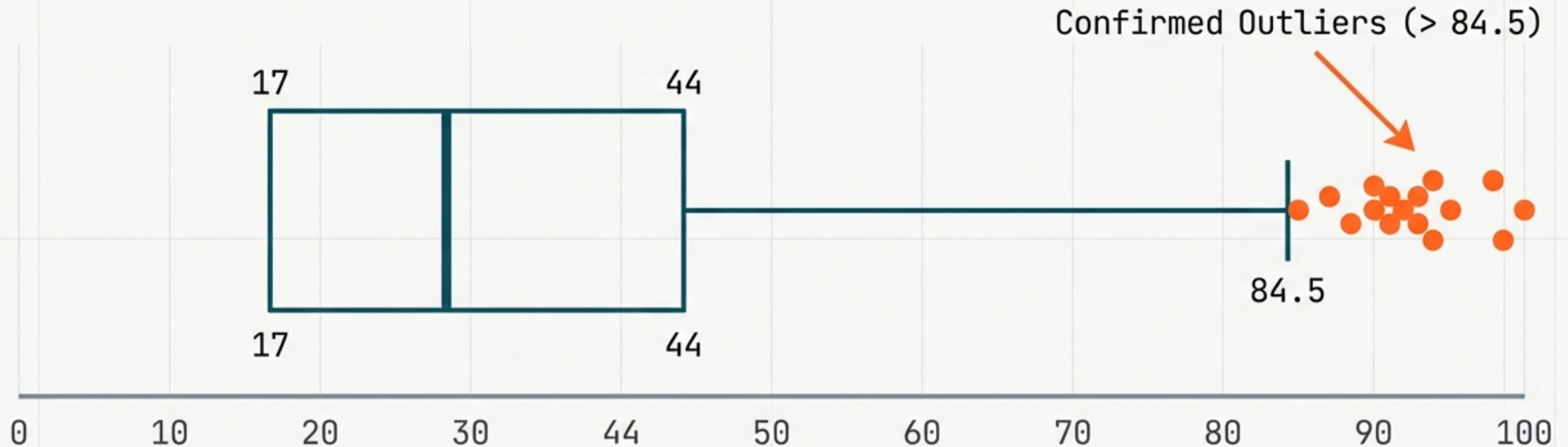


CRITICAL THRESHOLD:
We are looking for students
scoring above 84.5.

$$\text{Lower Limit} = 17 - (1.5 \times 27) = -23.5$$

~~Ignored: Marks cannot be negative.~~

Visualizing the Outliers



The plot confirms the diagnosis. We have valid data up to 84.5, and a cluster of extreme values beyond that. We now face a choice: Treatment A (Trimming) or Treatment B (Capping).

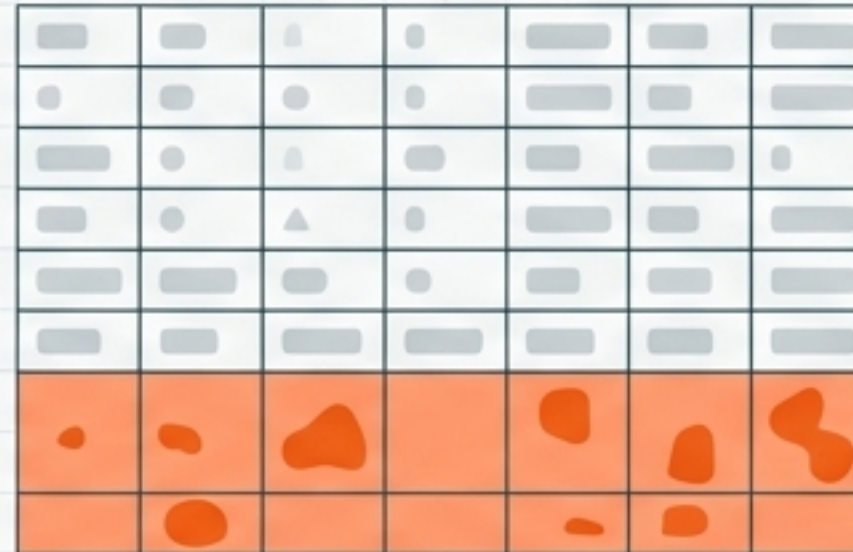
Treatment A: Trimming

The Logic

Removing the rows that contain outlier values entirely. This is “surgical removal”.

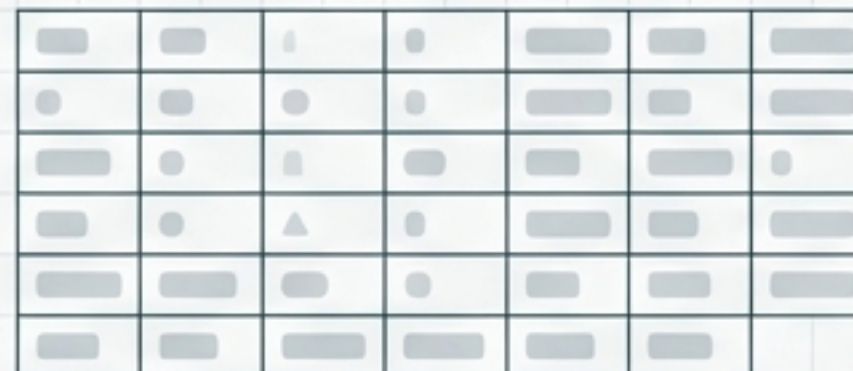
```
new_df = df[df['marks'] < 84.5]
```

Original Data (1000 rows)



↓ TRIM

Clean Data (985 rows)



Outliers Removed: **15**

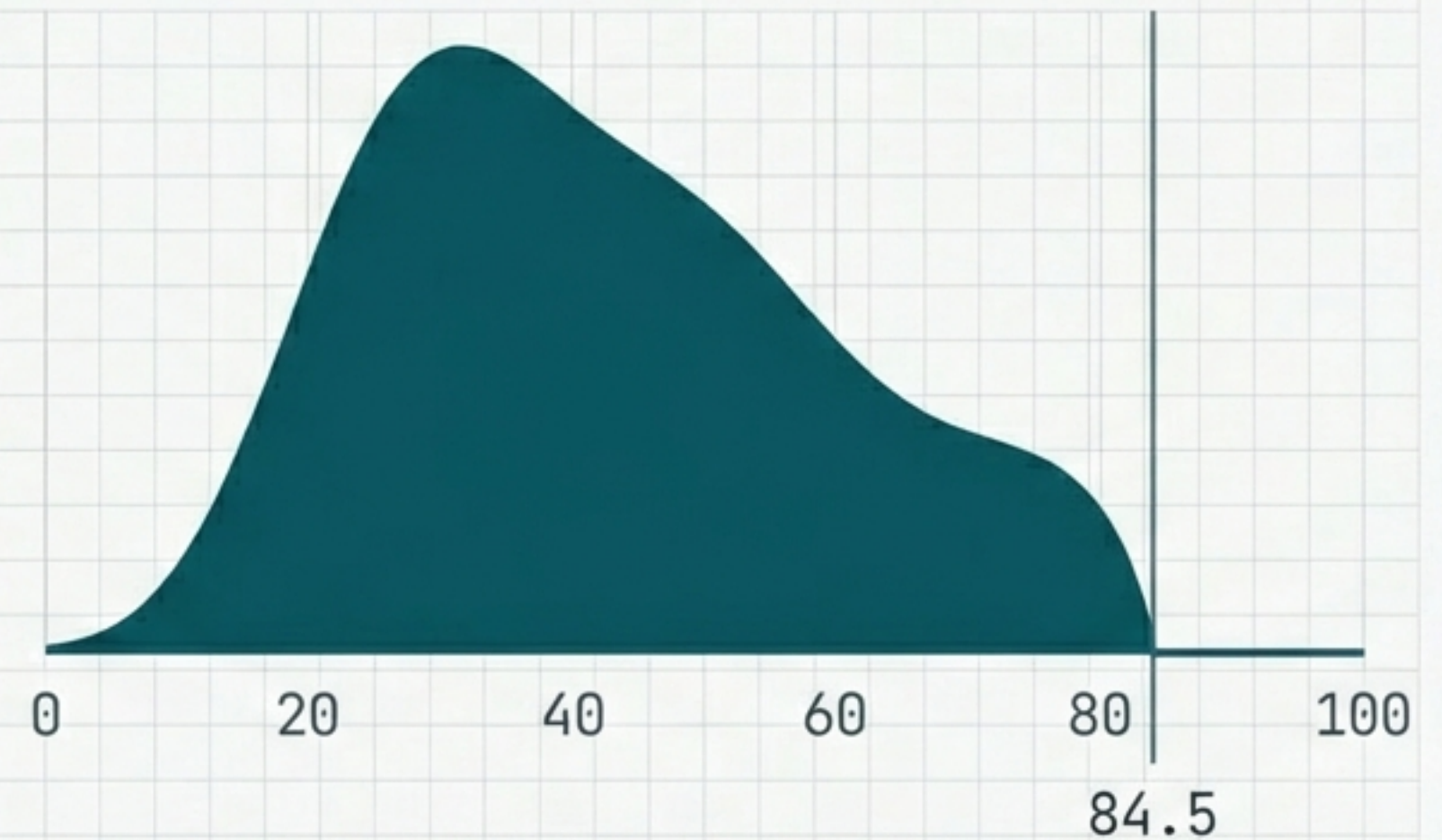
Data Loss: 1.5%

Trimming Results: Before vs. After

Before Trimming



After Trimming

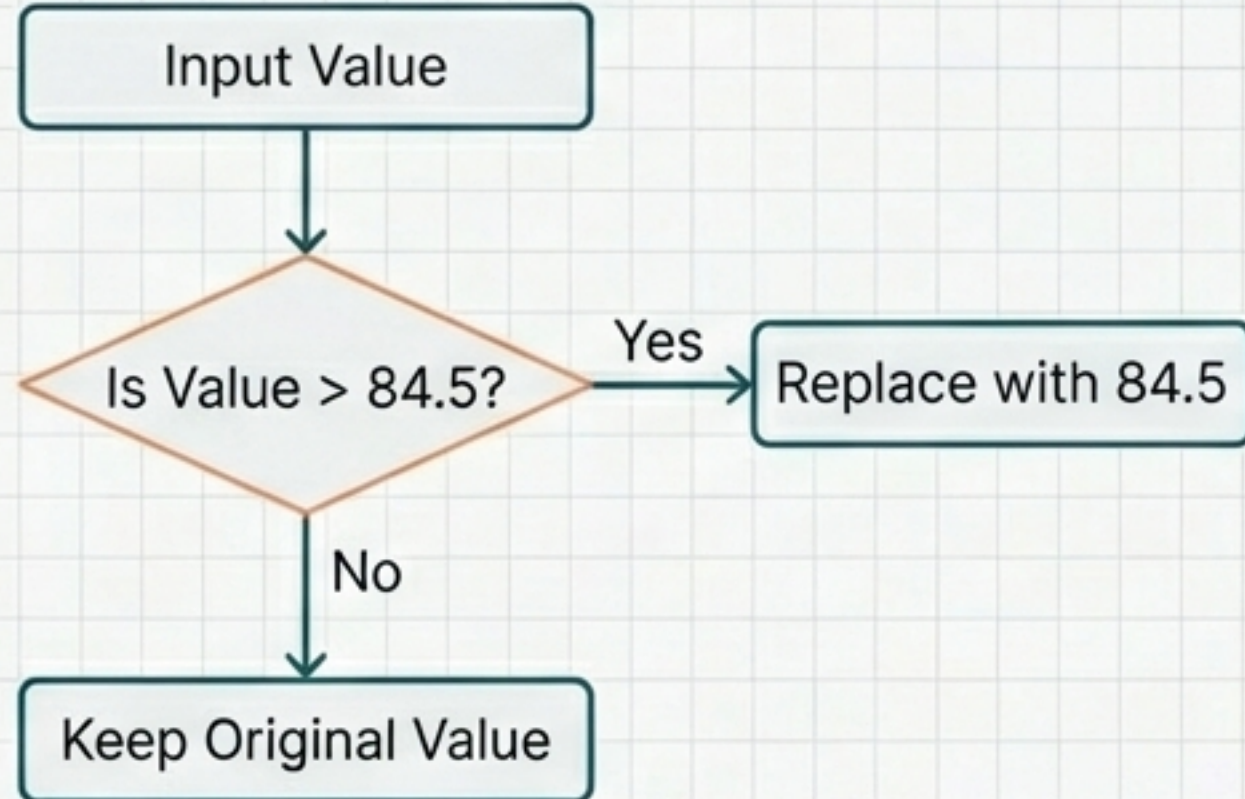


Result: The distribution is flatter. The extreme skew is gone, but the total volume of data has decreased.

Treatment B: Capping (Winsorization)

The Logic

Replacing extreme values with the limit thresholds. No data is lost; values are just “capped”.



Impact Assessment

Original Data (1000 rows)

●	●	●	●	▬	▬	▬
▬	●	▬	▬	▬	▬	▬
▬	●	▲	●	98	▬	▬
▬	▬	▬	●	▬	▬	▬
▬	▬	▬	▬	▬	▬	▬
▬	▬	▬	▬	▬	▬	▬

↓ CAP

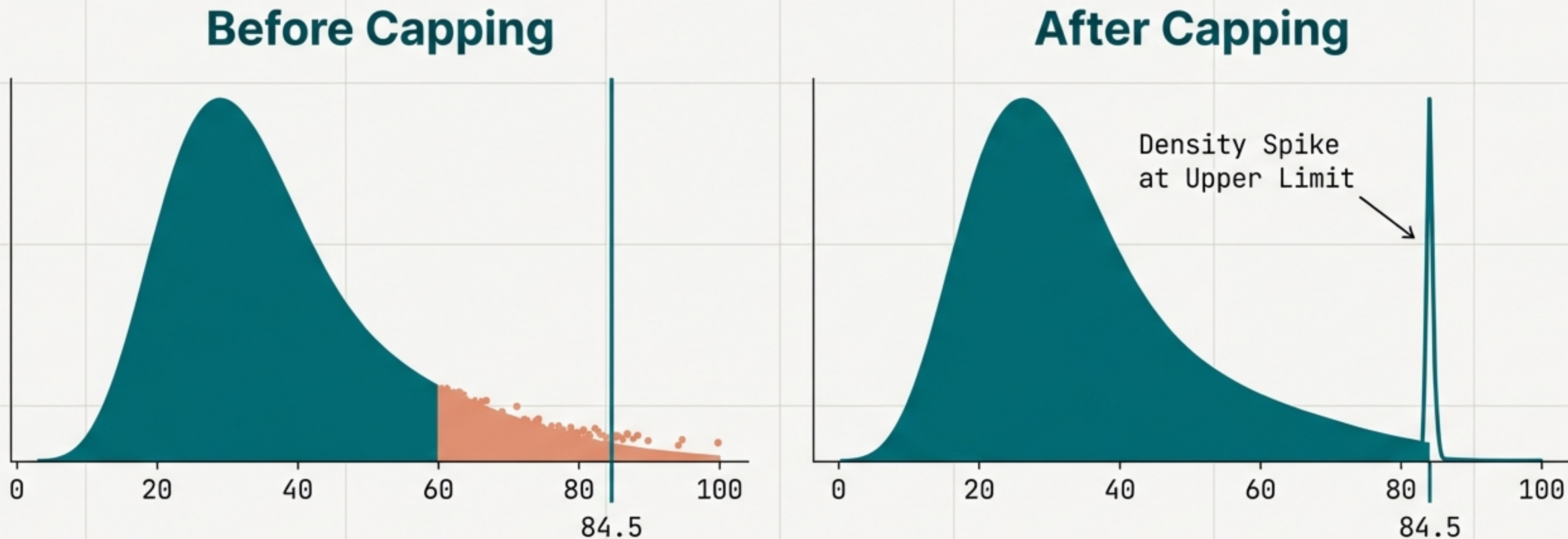
Clean Data (1000 rows)

▬	▬	▬	●	▬	▬	▬
●	▬	●	●	▬	▬	▬
▬	●	▬	▬	84.5	▬	▬
▬	▬	▲	●	▬	▬	▬
▬	▬	▬	▬	▬	▬	▬
▬	▬	▬	▬	▬	▬	▬

Outliers Modified: 15

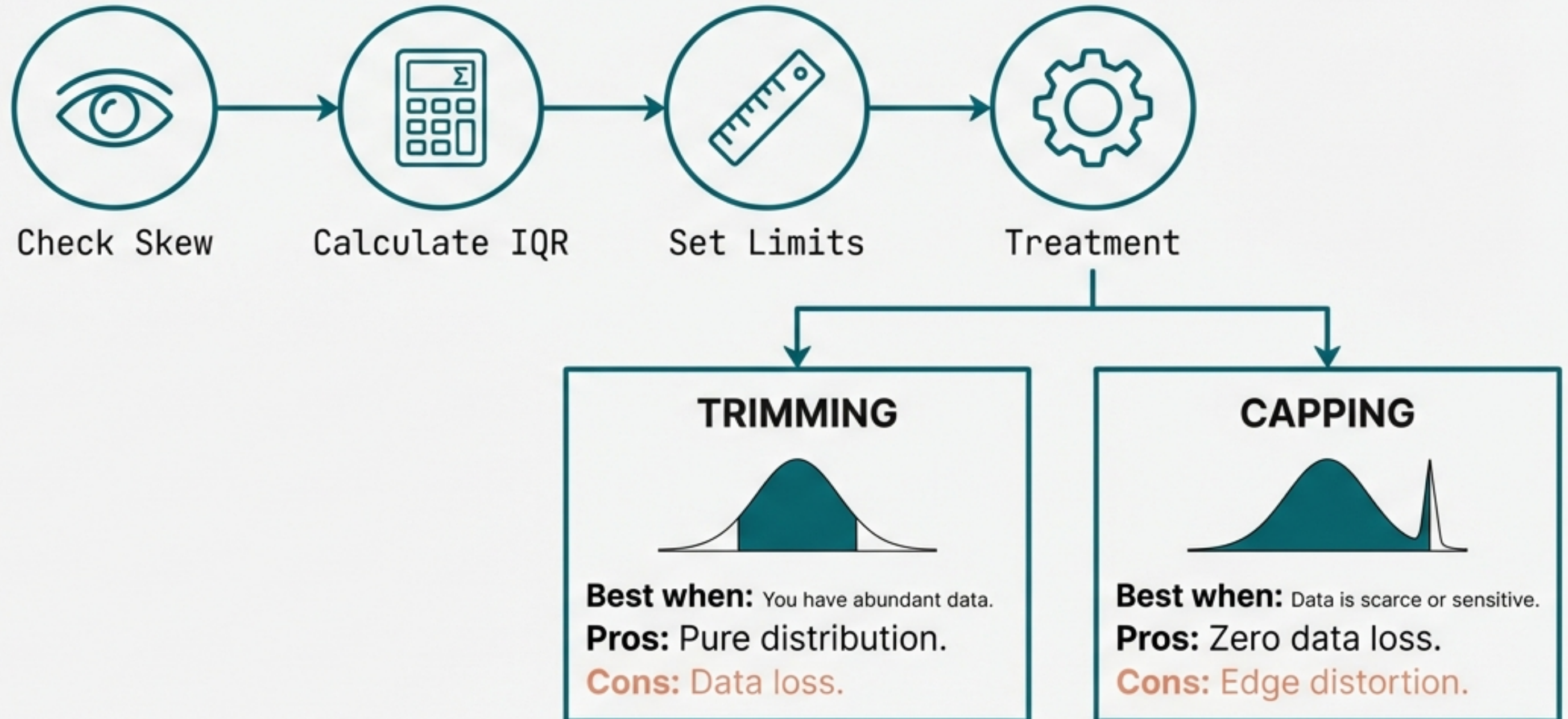
Data Loss: 0%

Capping Results: The 'Wall' Effect



Result: Outliers are shifted to the boundary. Distribution shape is preserved, but density at the edge is artificially increased.

Summary & Strategic Choice



The IQR method turns subjective outlier detection into an objective, repeatable mathematical process.